Exam in Signals and Transforms (1TE746)

January 12, 2026

Instructions

General:
- Use a separate page for each question and number the pages;
- Write your LADOK anonymization code on each page;
- **Clearly motivate your answers, missing motivations will give deductions;**
- Always simplify your answer as much as possible;
- Clearly mark your final answers (e.g., using double underlines or a frame);
- Always provide the correct unit, missing units will give deductions;
- Make sure that your calculator is set to radians.

Allowed Aids:
- The course's formula sheet;
- Template for Bode diagrams;
- One (1) single-sided A4 page with your own, hand-written notes on Chapters 1–5;
- One (1) single-sided A4 page with your own, hand-written notes on Chapters 6–9;
- Calculator;
- **No old exams, exercises, lecture notes, or other aids not listed above!**

Grading: The grade is based on the total score. The grading scale is as follows:
- Grade 3: 20–26.5 points *and* at least 50 % of the points in 5 out of the 6 questions.
- Grade 4: 27–33.5 points *and* at least 50 % of the points in 5 out of the 6 questions.
- Grade 5: 34–40 points *and* at least 50 % of the points in 5 out of the 6 questions.

Question 1: Continuous-time transform methods (8p)

Consider the spectrum of the signal $x(t)$ in Figure 1. Assume that the spectrum is zero outside of the shown interval and note that the phase spectrum is zero.

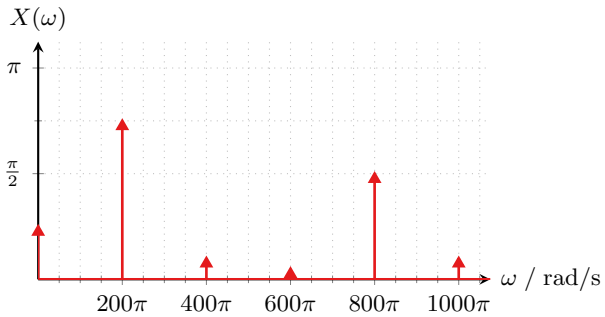

Figure 1: Spectrum of the signal $x(t)$ in Question 1.

a) (2p) Classify the signal in terms of its elementary signal properties. Name at least two elementary properties and motivate your answer.

b) (2p) Explain how the signal $x(t)$ looks in the time-domain (without explicitly determining the expression for $x(t)$). Use sketches to support your explanation (if helpful).

c) (2p) Let

$$y(t) = x(t) * 300\,\mathrm{sinc}(300t)$$

Sketch the signal $y(t)$ for the interval $0\,\mathrm{s} \leq t \leq 0.02\,\mathrm{s}$.

d) (2p) Assume that you are given an ordinary differential equation of the form

$$a_N \frac{\mathrm{d}^N z(t)}{\mathrm{d}t^N} + \cdots + a_1 \frac{\mathrm{d}z(t)}{\mathrm{d}t} + a_0 z(t) = b_0 v(t) + \frac{\mathrm{d}v(t)}{\mathrm{d}t} + \cdots + \frac{\mathrm{d}^M v(t)}{\mathrm{d}t^M}$$

Briefly explain how the Laplace transform can help in solving such differential equations.

Question 2: Continuous-time LTI systems (8p)

Consider the system with transfer function

$$H(s) = \frac{4}{s^2 + 0.2s + 2}$$

a) (2p) Determine the system's impulse response.

b) (2p) Assume that the input signal $x(t)$ is bounded in amplitude, that is, $|x(t)| \leq M_x$. Is the system's output signal $y(t)$ bounded in amplitude for any such bounded input $x(t)$?

c) (2p) Sketch the system's Bode magnitude diagram.

d) (2p) Assume that the input to the system is $x(t) = u(t)$. What is the output $y(t)$?

Question 3: Filtering theory (4p)

a) (2p) Consider the following lowpass filter requirements:

- Passband edge frequency: $\omega_{\text{pass}} = 2500\,\text{rad/s}$,
- passband ripple: $A_{\text{pass,dB}} = 1\,\text{dB}$,
- stopband edge frequency: $\omega_{\text{stop}} = 8000\,\text{rad/s}$,
- stopband attenuation: $A_{\text{stop,dB}} = 50\,\text{dB}$.

Determine the minimum order of a Chebyshev II filter fulfilling these requirements.

b) (2p) Real analog filters are typically built using electronic components. What challenges may this pose? Name and motivate at least two challenges.

Question 4: Discrete-time transform methods (8p)

Figure 2 shows the spectrum $X(\Omega)$ for the interval $0 \leq \Omega < 2\pi$ for a discrete-time signal $x[k]$.

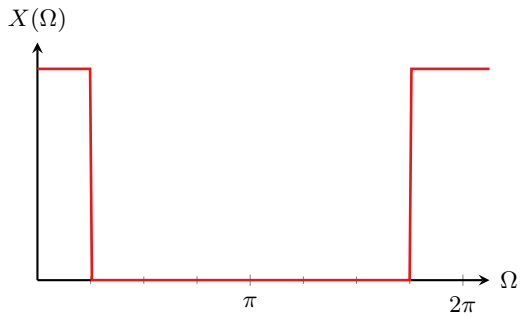

Figure 2: Spectrum of the signal $x[k]$ in Question 4.

a) (2p) Assume that the signal $x[k]$ (with spectrum $X(\Omega)$ in Figure 2) is obtained by sampling a continuous-time signal $x(t)$ with sampling frequency $\omega_s = 100\,\text{rad/s}$. Sketch the spectrum $X(\omega)$ of the signal $x(t)$ for the interval $-150\,\text{rad/s} \leq \omega < 150\,\text{rad/s}$. Motivate your answer.

b) (2p) Sketch the spectrum $Y(\Omega)$ of the signal

$$y[k] = x[k] \cos\left(\frac{\pi}{2}k\right)$$

for the inteval $-2\pi \leq \Omega \leq 2\pi$.

c) (2p) Assume that you are only given a finite number of K samples of the signal $x[k]$. How many samples K would you need to get the best possible result when calculating the spectrum of $x[k]$ using the discrete Fourier transform (DFT)?

d) (2p) *Windowing* can be used to affect the result of the discrete Fourier transform (DFT). What exactly can be improved by using particular windows (compared to just time limitation) and what does this mean for spectral analysis? What is the disadvantage of using windowing?

Question 5: Discrete-time LTI systems (8p)

Consider the discrete-time LTI system with transfer function

$$H(z) = \frac{1 + \sqrt{2}z^{-1} + z^{-2}}{1 - 0.4z^{-1} + 0.2z^{-2}}.$$

a) (2p) What is the difference equation describing the system's input-output relationship?

b) (2p) Sketch the system's pole-zero map. Is the system stable? Motivate your answer.

c) (2p) What are the system's gain and phase shift for $\Omega = \frac{\pi}{4}$?

d) (2p) Determine the output $y[k]$ of the system if the input is

$$x[k] = \sin\left(\frac{3\pi}{4}k\right) u[k].$$

Question 6: Sampling and reconstruction (4p)

a) (2p) Consider the spectra for the signals $x(t)$ and $y(t)$ shown in Figure 3. Assume that you are sampling the signal

$$z(t) = x(t) + y(t)$$

but that you are only interested in the component $x(t)$.

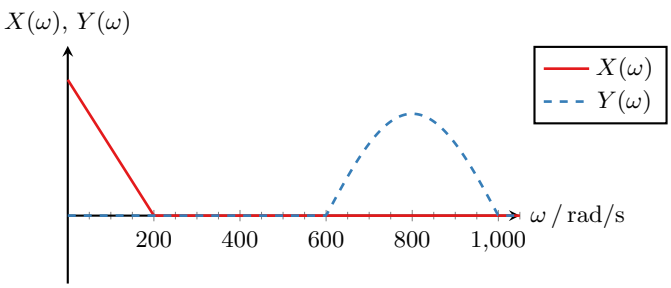

Figure 3: Spectra of the signals $x(t)$ and $y(t)$ in Question 6.

Give reasonable choices of pass- and stopband edge frequencies for a suitable anti-aliasing filter as well as the choice of sampling frequency ω_s for this case. Motivate your answers.

b) (2p) When sampling a continuous-time signal, an effect called *aliasing* may occur. What is aliasing and under what circumstances does this happen?